

Predictive Integrity & Wear Analysis

Hierarchical Bayesian Modelling for Mechanical Components

Technical Consultancy Project Report

Industrial Reliability & Failure Risk Assessment

Methodology: Hierarchical Bayesian Modelling (NumPyro / NUTS)
Focus Areas: Time-Series Degradation, Sparse-Data Prediction, Causal
Factor Identification

Portfolio Case Study

September 6, 2026

Client Brief: Component Reliability Assessment

Executive Background

An engineering client operates heavy machinery relying on a critical mechanical component (**Component K**). Component K experiences operational wear and tear over time, which is tracked via a non-destructive **Integrity Percentage** ($y_i \in [0, 100\%]$).

When a component's integrity drops to 30% or lower, it reaches failure threshold and must undergo refurbishment or replacement. The client requires a predictive framework to schedule proactive maintenance, identify physical wear drivers, and manage predictive uncertainty.

Problem Statement & Client Objectives

The client has engaged an independent consultancy to solve three primary operational challenges:

1. **Degradation Characterisation:** Accurately model and characterise individual degradation rates over time based on historical time-series data.
2. **Sparse Data Prediction:** Predict future integrity and compute probability of failure ($y_i \leq 30\%$) at day $t = 30$ for newly deployed components with limited initial observations ($t \leq 10$).
3. **Causal Analysis of Wear:** Determine which physical characteristics (X_1 through X_5) drive degradation to inform targeted engineering interventions.

Supplied Data Overview

The client provided time-series measurement data across 75 individual components:

- **Dense Historical Set (Indices 0–49):** 50 components with regular, well-populated integrity readings across up to 45 operational days.
- **Sparse Test Set (Indices 50–74):** 25 newly introduced components with sparse readings (1 to 9 measurements) gathered exclusively within the first 10 days ($t \leq 10$).
- **Static Characteristics (X_1 – X_5):** Five real-valued physical measurements static to each component.

Modeling Expectations & Constraints

Engineering Challenges & Methodological Directives

Standard deterministic machine learning approaches (e.g., standard neural networks, SVRs) are unsuited to this operational setting due to irregular observation intervals, severe data sparsity in early deployment, and the requirement for explicit probabilistic risk bounds[cite: 1, 2].

The client specifically mandates a **Hierarchical Bayesian framework** (implemented via Hamiltonian Monte Carlo / NUTS sampling) to enable partial pooling—allowing sparse components to borrow statistical strength from densely observed historical units[cite: 1, 2].

Model Variant	Functional Form & Description
Baseline Model	$f_i(t) = u_i \exp(-v_i t/100)$ Exponential decay parameterised by initial integrity $u_i \in [80, 100]$ and baseline decay rate $v_i \in [1, 10]$.
Enhanced Model	$f_i(t) = u_i \exp\left(-\left(v_i + \sum_{j=1}^5 w_j x_{ji}\right) t/100\right)$ Incorporates physical features X_1 – X_5 via shared weight parameters w_j to explain variation in component wear rates.

Table 1: Formulations specified in the client requirement.

Contents

1	Modelling Framework	2
1.1	Baseline Model	2
1.2	Enhanced Model	3
1.3	Prior specification	3
1.4	Running and Diagnostics	4
2	Predictive Modelling	5
2.1	Baseline Results	5
2.2	Enhanced Results	8
2.3	Comparison	10
3	Potential Causes of Degradation	11
3.1	Weights	11
3.2	Recommendations	12
4	A Black-box Model	12
4.1	Black-box Model	12
4.2	Model Performance	12
5	The Support Vector Machine	14
6	Conclusions	14

1 Modelling Framework

The client operates machinery where components lose integrity y_i over time and need maintenance when $y_i < 30\%$. They have supplied data from 75 components, the first 50 are well-populated with data up to day 30, and the later 25 are sparsely populated up to day 10.

This report has three objectives, model the degradation curve, predict future integrity and probability of maintenance (particularly for data-poor components) and identify characteristics influencing degradation.

To meet these objectives a hierarchical Bayesian approach will be taken.

1.1 Baseline Model

The brief states that integrity measurements (y_i) for component i at day t follow a noisy exponential decay:

$$\begin{aligned} y_i(t) &= f_i(t) + \epsilon, \text{ with,} \\ f_i(t) &= u_i \exp(-v_i t / 100), \end{aligned} \tag{1}$$

Where u_i is the starting integrity and v_i is the decay rate.

Our hierarchical model is defined with three levels as shown in Figure 1. Within our model this equation 1 is evaluated at each observation time-point and used as the mean of a Student-t likelihood.

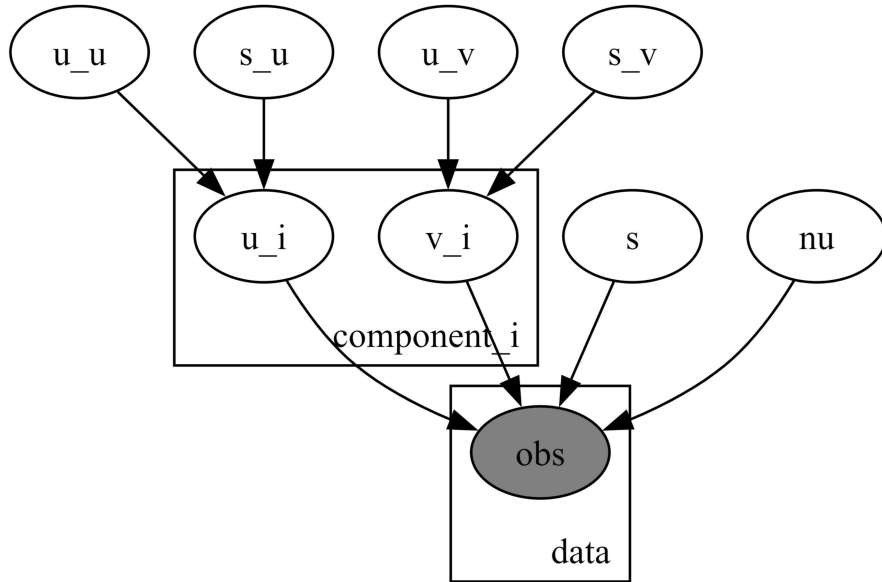


Figure 1: Baseline Model Structure

The component parameters u_i, v_i are drawn from population-level distributions with means μ_u, μ_v and scales σ_u, σ_v which are inferred from the data. This "partial pooling" allows sparse components borrow strength from well-observed ones while still retaining individual differences.

The scaling factor ν has been included after running PSIS-LOO diagnostics which flagged certain data points having Pareto $k > 0.7$. This indicates that outliers within the data were having disproportionate influence on the importance weights. The Gaussian error term was replaced by a Student-t distribution where the degrees-of-freedom parameter ν is learnt from the data.

1.2 Enhanced Model

To capture possible physical drivers of wear, we extend the exponent with the five covariates x_{ji} :

$$f_i(t) = u_i \exp \left\{ - \left(v_i + \sum_{j=1}^5 w_j x_{ji} \right) t / 100 \right\} \quad (2)$$

The enhanced model adds $w^T x_i$ to v_i then applies the resulting decay rate as in the baseline.

The weights w_j are shared across all components and learn for each characteristic x_j s.

Figure 2 shows the extended model structure

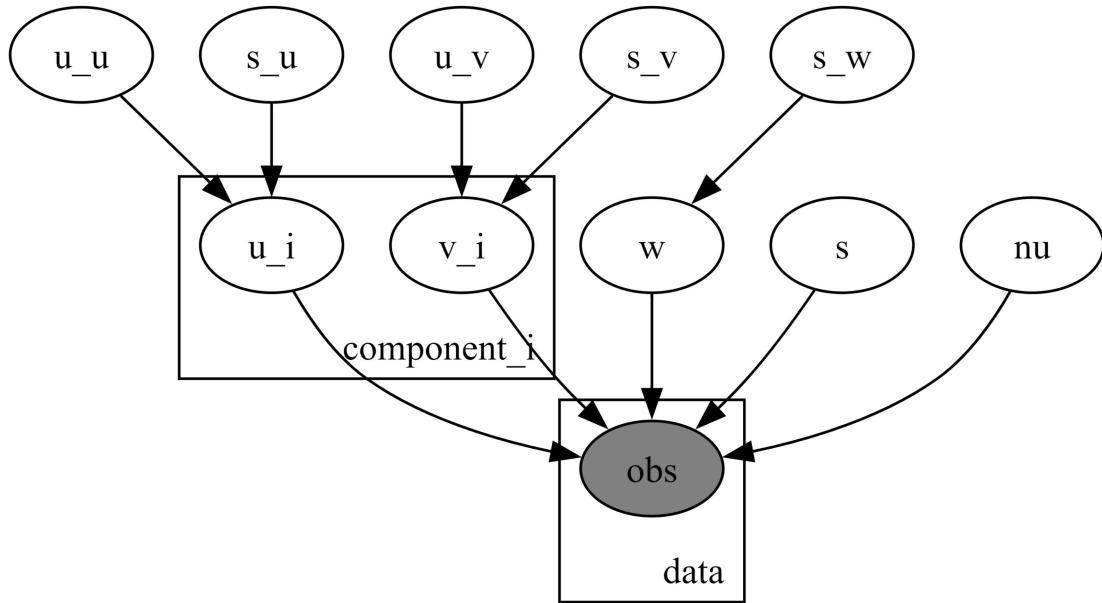


Figure 2: Enhanced Model Structure

1.3 Prior specification

Within our models, we define our priors based upon the brief supplied by the client. They are defined as follows:

Parameter(s)	Prior	Justification
$\sigma_u, \sigma_v, \sigma_w^\dagger, \sigma$	Half-Normal(3)	Half-Normal ensures scales are non-negative; scale 3 is weakly informative, allowing the model to fit to the data.
u_u	TruncNormal(90, 5) [0, 100]	Client expects initial integrity 80–100 %; centred at 90 with s.d. 5, truncated to valid range.
u_v	TruncNormal(3, 2) [0, ∞)	Client expects decay 1–10; trace plots show most mass near 3, so centre shifted from 5; truncation stops negative decay.
u_i	TruncNormal(u_u, σ_u) [0, 100]	Component intercepts pooled around u_u with spread σ_u ; truncation keeps values on 0–100 scale.
v_i	TruncNormal(u_v, σ_v) [0, ∞)	Component decay rates pooled around u_v ; truncation enforces non-negative wear.
ν	Exponential(3)	Puts most mass on small ν , giving heavy tails to curb outliers, yet lets the posterior move higher if data are clean; chosen so Pareto- $k < 0.7$.
$w^\dagger$	Normal(0, σ_w)	No prior engineering bias across diagnostics; mean-zero, scale σ_w (learned) keeps weights weakly informative.

† Enhanced model only.

Table 1: Prior distributions and their rationale.

1.4 Running and Diagnostics

Both models were ran with NumPyro’s No-U-Turn Sampler (NUTS) with 4 chains, 2000 warm-up steps and 2000 samples, yielding 8000 posterior samples. Diagnostics are displayed in Table 2.

Model	$\hat{r}_{\max}$	Divergences	Parteo $k > 0.7$	min ESS _{bulk}	min ESS _{tail}	Runtime
Baseline	1.00	0	0	1892	3139	65 s
Enhanced	1.01	0	0	406	693	89 s

Table 2: Sampling diagnostics.

The maximum $\hat{r}$ value was 1.01 with 0 divergences, indicating both models converged to the target distribution. The PSIS-LOO-CV Pareto- $k > 0.7$ count was 0 for both models indicating that no observations exceeded the reliability threshold and the heavy-tailed Student-t error was able to limit high-influence outliers.

The minimum number of effective samples indicates how many independent samples drawn. Vehtari et al. (2021) recommends a minimum ESS of 400 for stable Monte-Carlo errors. For the baseline model, both the ESS_{bulk} (central mass) and the ESS_{tail} (extreme quantiles) are above this threshold. The enhanced model was just above 400, indicating that running longer chains would have minimal effect on the error precision.

The runtime for both models was under 2 minutes, making retraining practical when updating with new data.

2 Predictive Modelling

This section will discuss the results of NUTS sampler on our two hierarchical Bayesian models.

2.1 Baseline Results

Figure 3 and 4 show the baseline model’s trace plots with kernel densities (left) and MCMC traces (right).

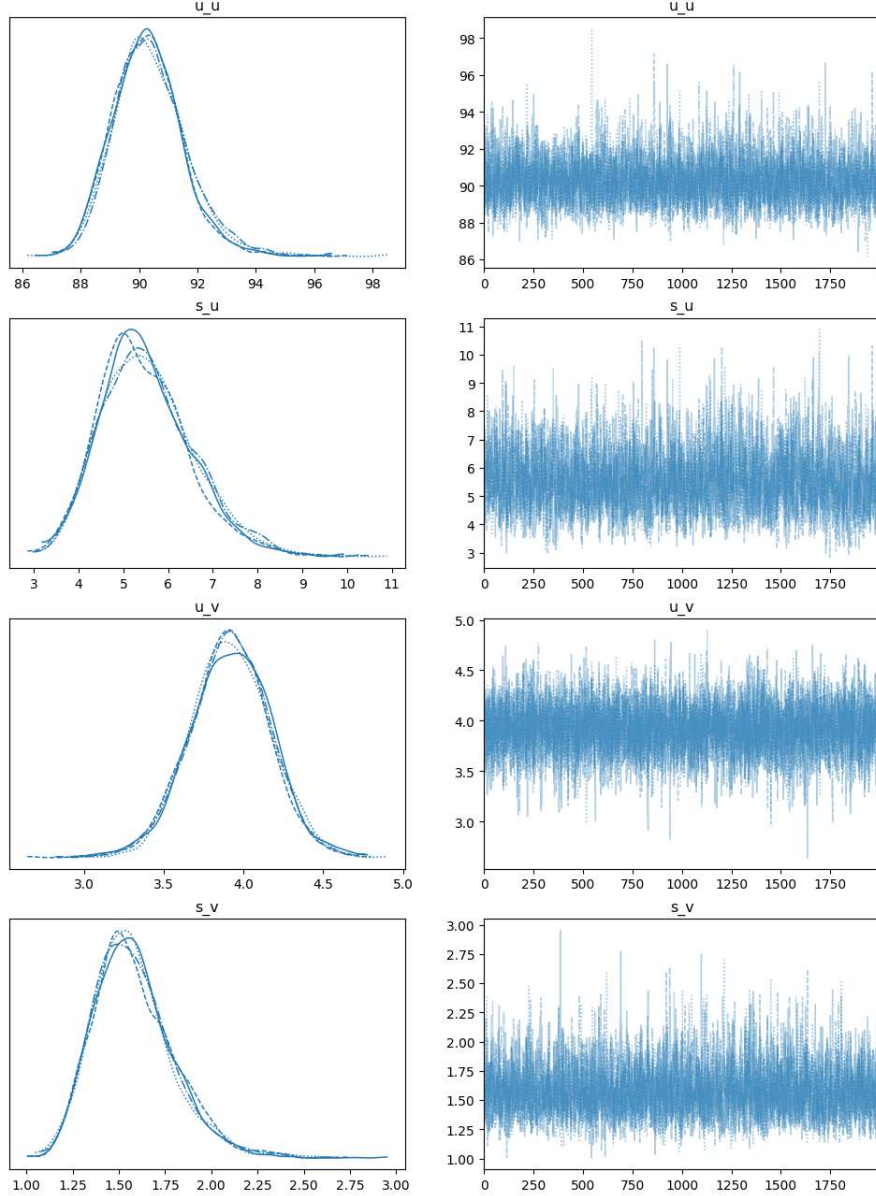


Figure 3: Baseline Trace Plot (1)

All four chains mixed well, with no drift visible in the MCMC traces. This is inline with the max $\hat{r} = 1.00$ and zero divergences (Table 2).

The mean initial integrity u_u sits in the centre of the client’s expected range of 80-100%, with variance s_u around 5.5 indicating, roughly 95% of components are in a ± 10 spread

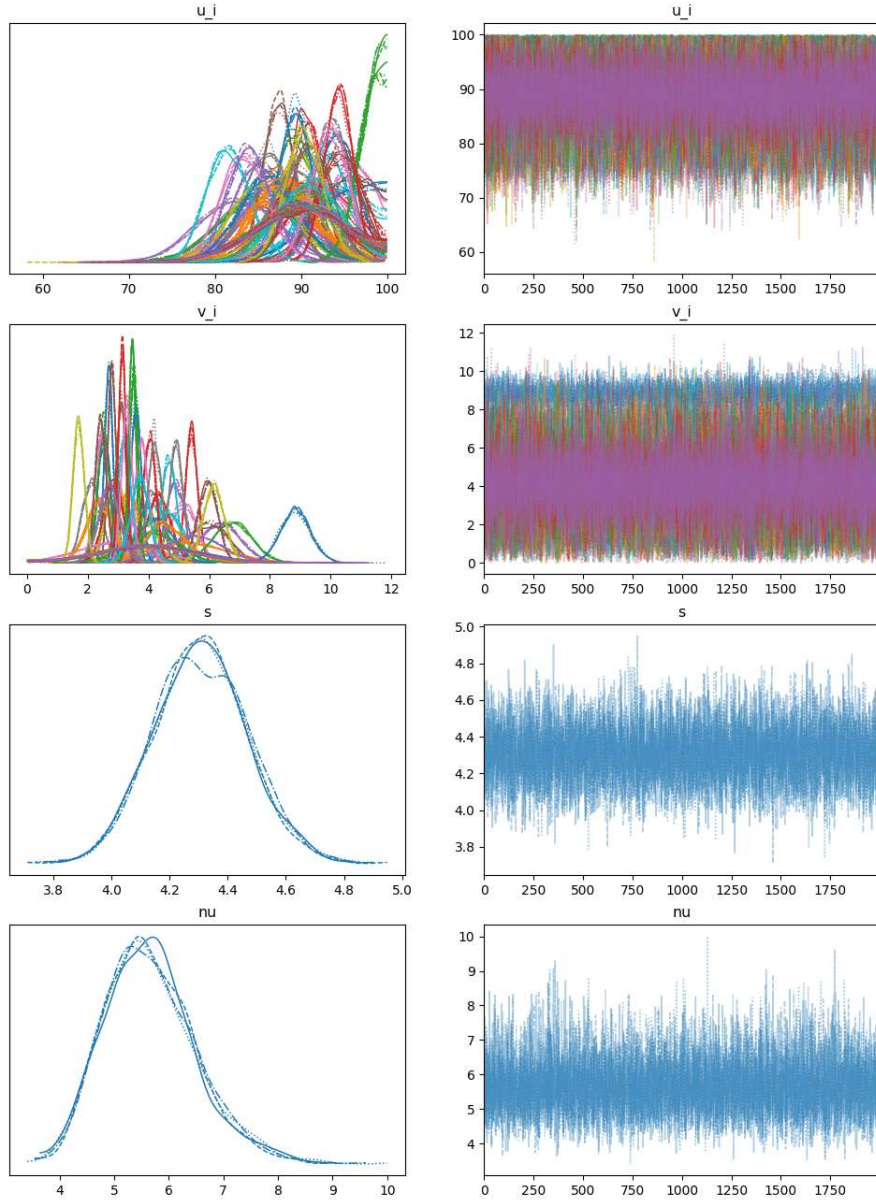


Figure 4: Baseline Trace Plot (2)

of the mean under a normal assumption. Some individual u_i distributions are truncated at 100, and some of the tails extending past the expected range and down into the 70% range.

The mean decay rate u_v is 3.9%, well within the expected 1-10% range. Individual decay distributions v_i indicate certain components exhibit faster decay rates ($< 6\%$), with one near 9%.

The Student-t degrees of freedom ν is centred around 5.7 suggesting the heavy-tailed likelihood has limited the effect of the outliers without destabilising the posterior.

Figure 5 displays the model fit for four components, showing the uncertainty of both the underlying function estimate and overall model with Table 3 displaying summary statistics.

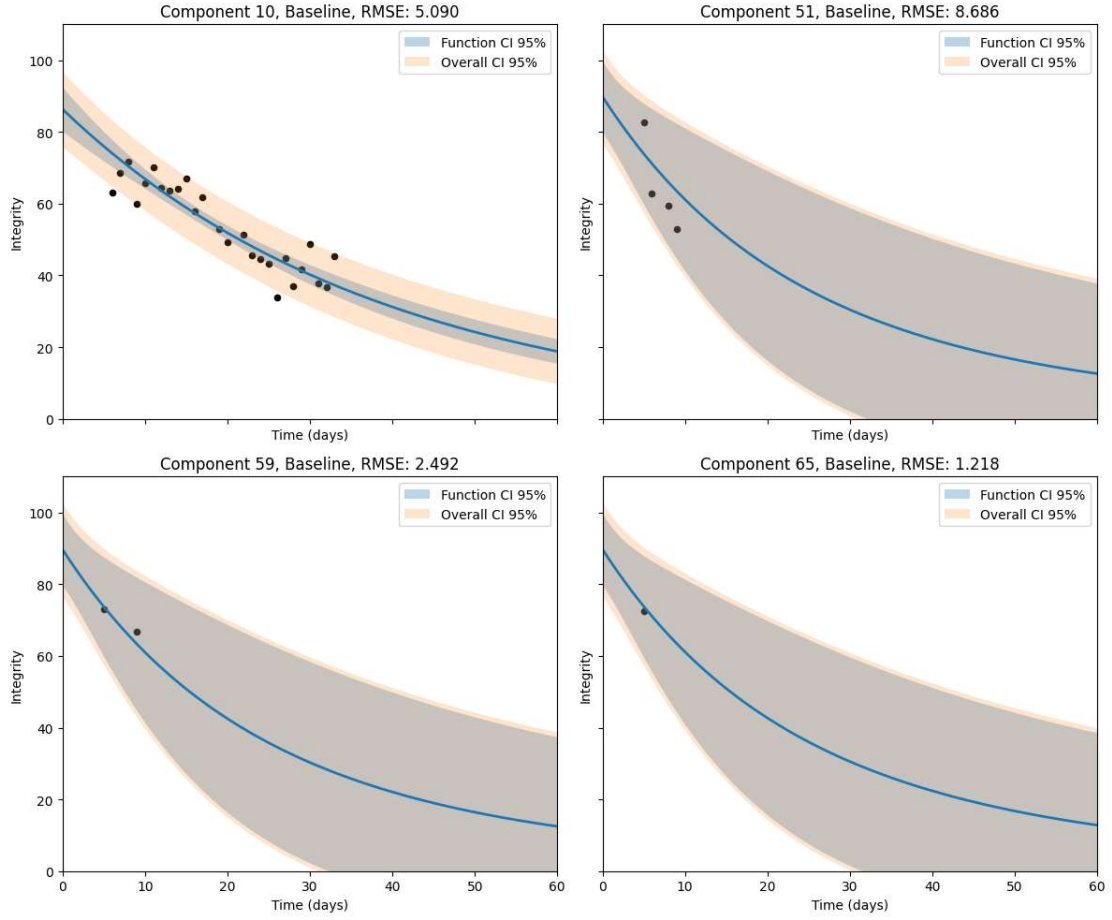


Figure 5: Baseline Prediction Plots

Component	RMSE	MAE	Coverage (%)	Mean Width
10	5.090	3.883	92.3	17.65
51	8.686	8.557	100.0	37.02
59	2.492	2.142	100.0	36.52
65	1.217	1.217	100.0	32.90

Table 3: Baseline Model Component Performance Metrics

Component 10 has many observations leading to u_{10} v_{10} being well defined, and producing a narrow 95% confidence interval (CI) (mean width 18%), implying predictions are within $\pm 9\%$ of the true value. With a low RMSE of 5.1 and overall uncertainty bands covering 88.5% of values, the model fits the densely sampled data accurately.

For sparse components (51, 59, 65) the model uses information from the dense components to produce reasonable decay curve with minimal data (e.g. 1 data point for component 65). However, after $t > 10$, predictive uncertainty grows significantly. At $t=30$, the model's CI spans roughly $2 < y_i < 60$, with mean widths of 33-37 for these 3 components, limiting the model's practical guidance for maintenance scheduling. Whilst RMSE and MAE values are low, the small number of evaluative points limits their reliability as performance metrics.

2.2 Enhanced Results

Figure 6 and 7 display the enhanced model's trace plots (excluding weight parameters).

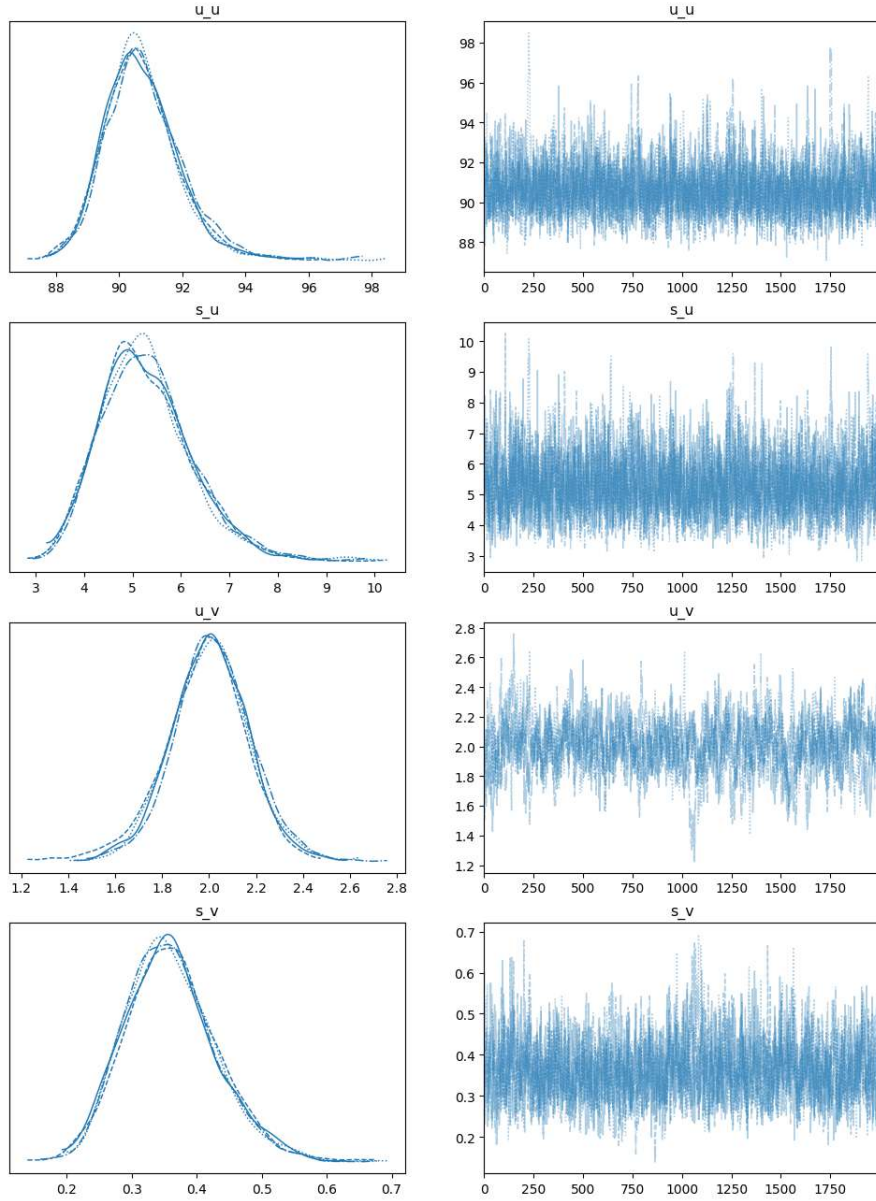


Figure 6: Enhanced Trace Plot (1)

These trace plots closely resemble those of the baseline model (Figure 3, 4). As before, the four chains mixed well with no visible drift, which is consistent with diagnostics from Table 2.

Kernel densities for the initial integrity parameters (u_u, s_u, u_i), observation variance (s) and degrees of freedom (ν) are similar to the baseline model.

The main difference is within the decay rate (v). Incorporating the five physical characteristics shifts the mean of u_v from 3.9 to 2.0 and reduces variance s_v from 1.58 to 0.36. This implies that these 5 characteristic now explain a lot of the variance in decay rate. The v_i traces are now more tightly centred around 2, with previous outliers (e.g. decay $\approx 9\%$ no longer present).

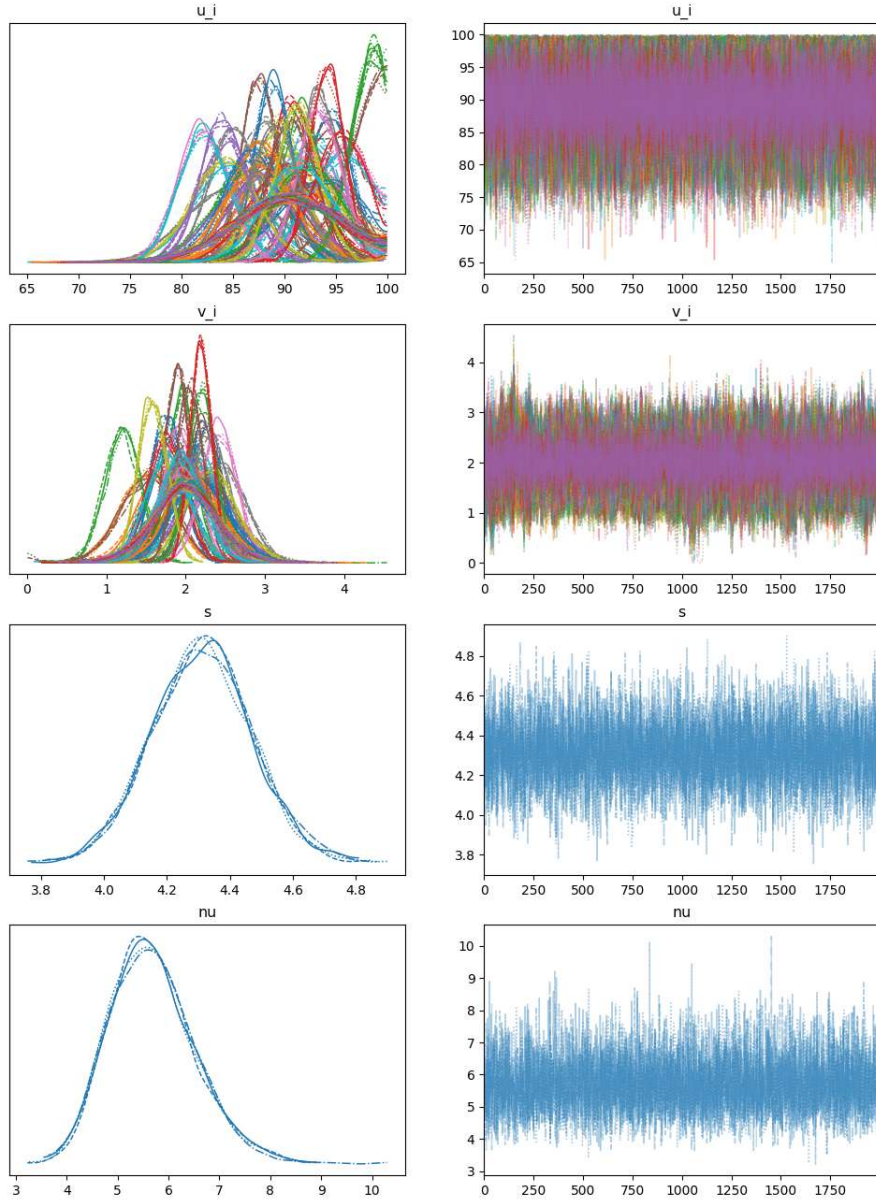


Figure 7: Enhanced Trace Plot (2)

This increased confidence in decay rate is evident in Figure 8. For the well-sampled component 10, performance is comparable to the baseline. However, for sparsely populated components, the mean width shrinks to between 22-25% (Table 4), indicating significantly greater certainty. Although these tighter confidence intervals cover slightly fewer observed data points than the baseline, the trade-off for increased precision is acceptable.

Component	RMSE	MAE	Coverage (%)	Mean Width
10	5.157	3.874	88.5	17.59
51	7.580	4.890	75.0	22.28
59	2.939	2.407	100.0	24.63
65	1.305	1.305	100.0	23.44

Table 4: Enhanced Model Component Performance Metrics

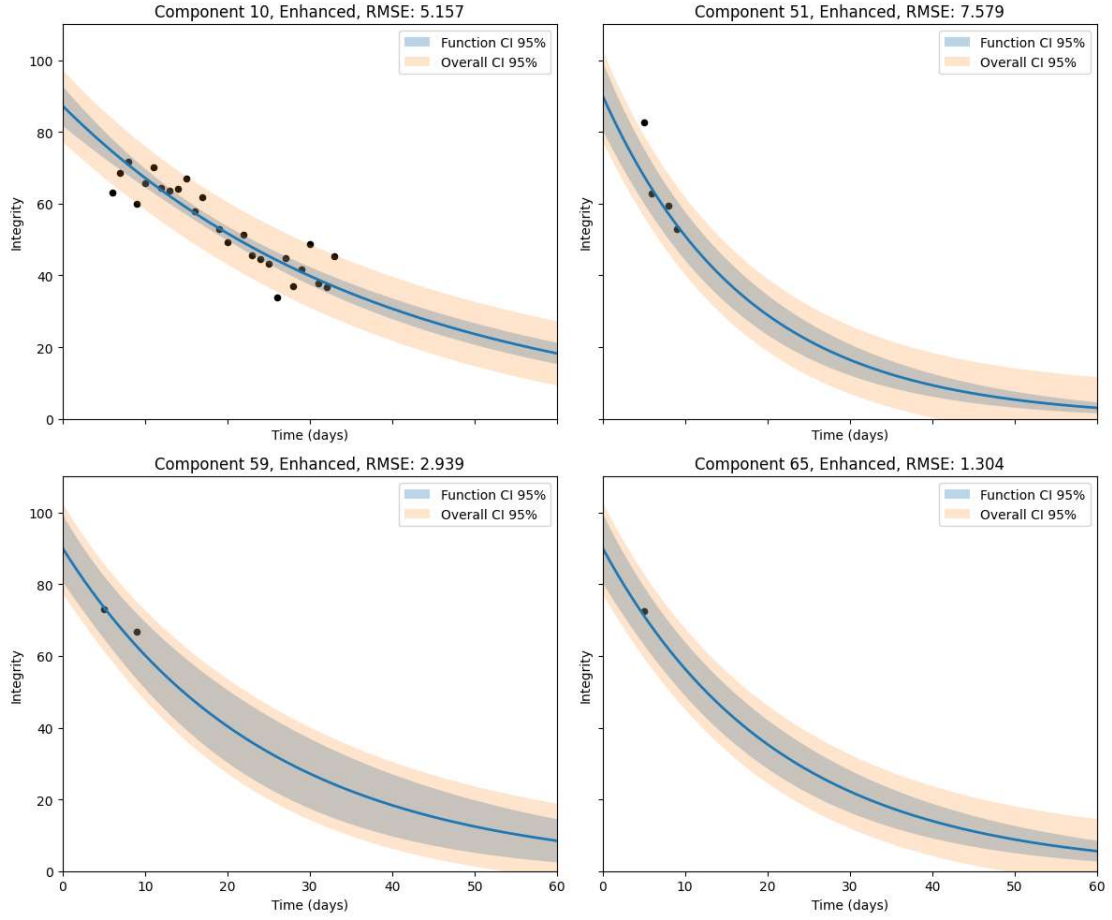


Figure 8: Enhanced Prediction Plots

2.3 Comparison

Table 5 summarises the overall metrics with both models showing strong performance with low MAE values. The enhanced model shows a 9% reduction in RMSE over the baseline which from Figure 5 and 8 is likely due to the higher certainty in sparse components.

Although the enhanced model has slightly lower coverage in the four example plots, overall coverage remains similar. Confidence bands shrink by about 1% (Table 3 and 4), which again is mainly driven by sparse components.

Pareto-smoothed importance-sampling leave-one-out (PSIS-LOO) provides an approximate, out-of-sample log-predictive score. The enhanced model has a slightly less negative expected log predictive density (elpd-loo), with similar standard error, indicating a marginal improvement over the baseline. It's effective number of parameters (p-loo) is also lower, indicating it achieves its predictive accuracy with less complexity. This is likely due to the inclusion of the 5 characteristic weights explaining more of the variance.

Given the enhanced model offers consistent marginal gains in predictive performance and tighter predictive bands, it is recommended and will be used in the blind-tests.

Model	RMSE	MAE	Coverage	Mean Width	elpd loo	elpd s.e.	p loo
Baseline	5.531	4.142	93.1	19.27	-2492	20.82	75.27
Enhanced	5.029	3.958	93.4	18.28	-2484	20.90	63.15

Table 5: Global Model Performance Metrics

3 Potential Causes of Degradation

This section inspects the diagnostic weights and considers the driving factors of degradation within the components.

3.1 Weights

Figure 9 shows the trace plot for σ_w and w , with Table 6 summarising these results. Positive weights accelerate degradation while negative weights slow it (Equation 2). The MCMC traces indicate that the chains mixed well and there is no visible drift which validates our results.

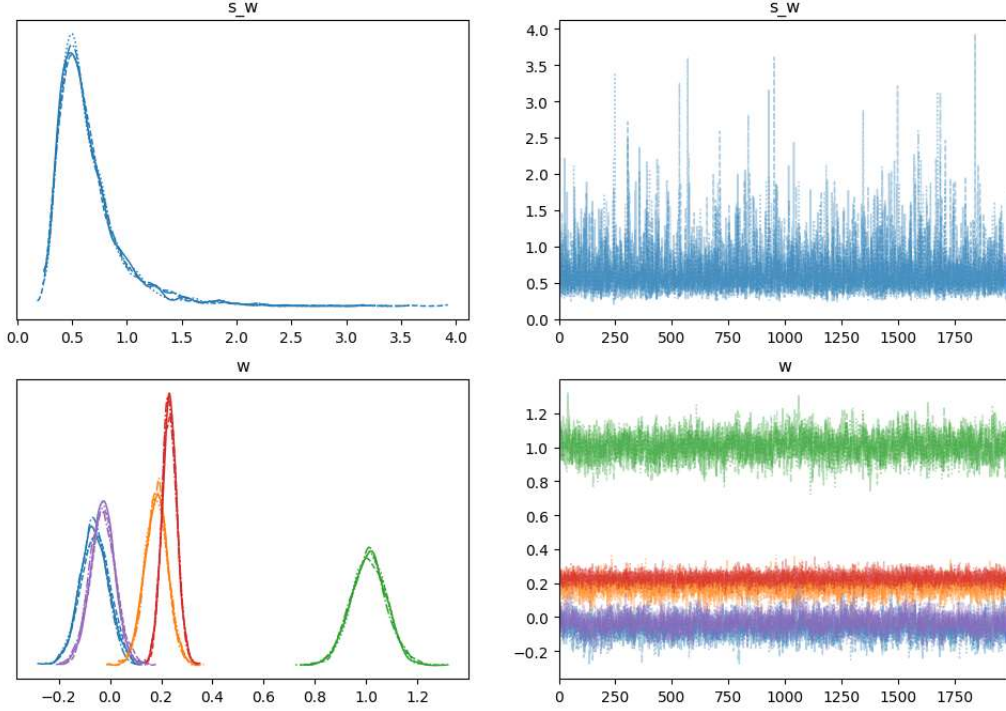


Figure 9: Enhanced Weight Trace Plot

Weight	Mean	SD
w[0]	-0.060	0.057
w[1]	0.180	0.046
w[2]	1.009	0.070
w[3]	0.232	0.030
w[4]	-0.029	0.049

Table 6: Weight Metrics

Table 6 indicates that X3 (w_2) is the main driver of degradation: each one-unit increase in X3 raises the effective decay rate by about 1% per 100 days. The base decay rate u_v (Figure 6) is around 2, and the average X3 value is ≈ 1.2 . Therefore, a two-unit increase above the average, raises the effective decay rate from roughly 3.2 to 5.2, reducing useful life by $\approx 40\%$.

Characteristics X2 and X4 also impact on degradation, each adding around 0.2% to the decay rate. Whilst small individually, the effect can compound - as in the case of component 66, which has $X2 = 10.04$ and $X4 = 12.41$.

In contrast, X1 and X5 weights are centred near 0, suggesting minimal impact on integrity decay.

3.2 Recommendations

Given this information, components with higher levels of X3 should be placed on a shortened inspection or replacement interval, and if resource are more tight prioritise components with both high levels of X3 and, X2 or X4. Characteristics X1 and X5 showed no significant link with degradation, so consider re-allocating resources from measuring X1 and X5 to potential methods to lower the impact of X3, saving money in reduced replacement and maintenance costs.

4 A Black-box Model

To address the engineering team’s question around using a more simplistic black-box model, this section outlines and assess a basic black-box Random-Forest Regressor comparing the results to our Bayesian model.

4.1 Black-box Model

The black-box model uses some summary statistics derived from data before $t=10$: mean integrity, standard deviation, least-squares slope and intercept, last observed value and log-mean. Alongside the static characteristic X1-X5, these form an 11-dimensional feature set.

The dataset was filtered to remove components ≤ 10 observations. If our target integrity at $t=30$ was missing, it was inferred from the Bayesian enhanced model. The resulting dataset was split 80/20 into training and test sets.

We use a standard Random Forest Regressor (SciKit-Learn) with 100 trees and no hyper-parameter tuning.

4.2 Model Performance

Figure 10 shows the test set predictions. The model achieved an RMSE of 8.193 but tended to underestimate integrity levels. The model trained in under 2s, suggesting larger, more complex black-box models, would also remain practical. However, compared with the enhanced Bayesian model’s RMSE of 5.029, the black-box model underperforms.

Predictions for four sparsely sampled components are overlaid with an ‘x’ on the enhanced Bayesian model’s predictions in Figure 11. For three components, black-box estimates fall

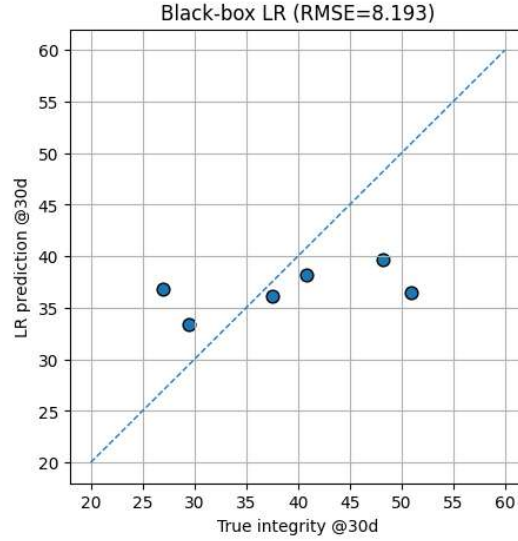


Figure 10: Black Box Predictions on Test Set

within the Bayesian model's 95% CI. However, it is unable to predict on component 65 as it only has one data point.

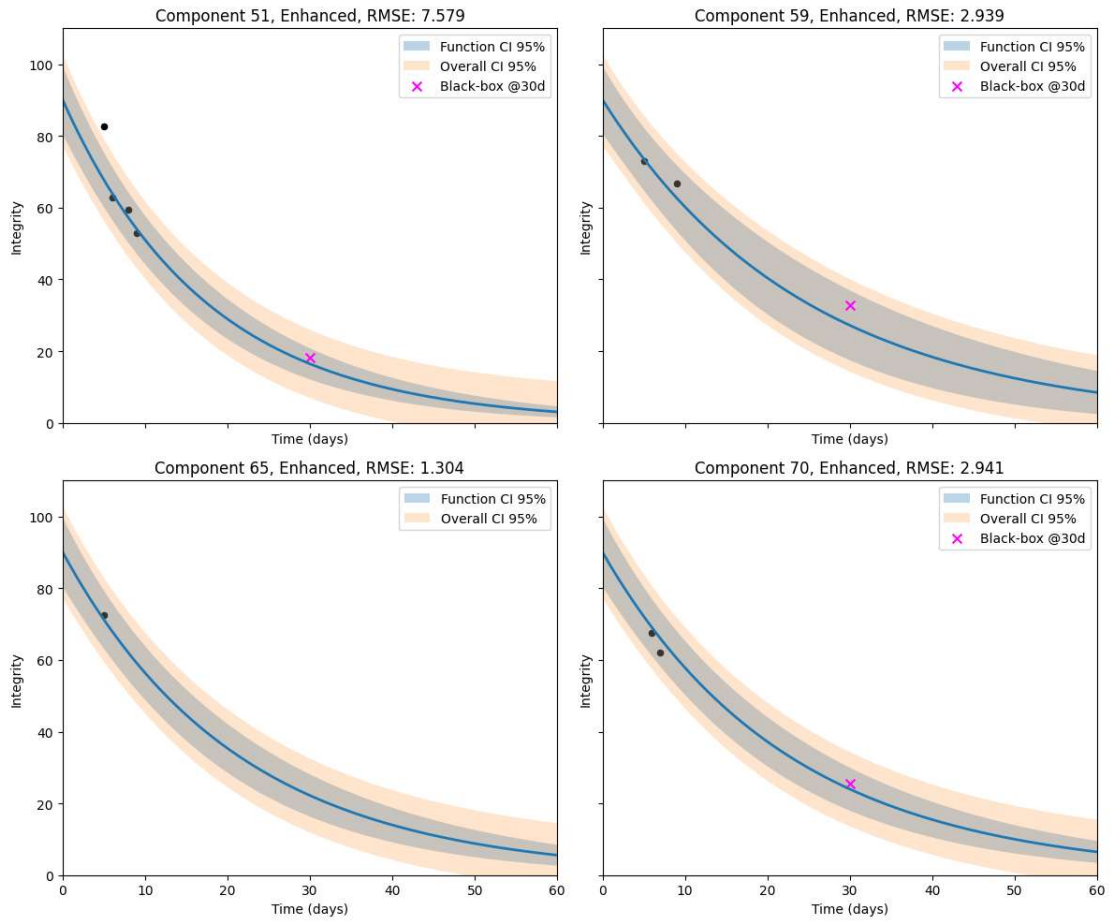


Figure 11: Black Box Predictions Compared With Enhanced Bayesian Model

Furthermore, the black-box model outputs only point estimates and doesn't consider

uncertainties within that prediction. Translating that into a probability of breaching the 30% threshold would require a separate calibration step which could be unreliable.

In comparison, the Bayesian model offers four primary advantages:

1. it handles irregular sparse time-series without imputation,
2. it enables encoding of known exponential dynamics, ensuring plausible long-term extrapolation,
3. it borrows strength from more populated components, tightening predictions for data-poor components and
4. it produces full predictive distributions, enabling direct probability estimation

5 The Support Vector Machine

A Support-Vector Regressor (SVR) projects the data into a high-dimensional feature space with a kernel and fits an ϵ -wide tube, balancing flatness and the penalties for errors beyond ϵ (Drucker et al., 1997). The solution depends on a small number of support vectors, making predictions fast on large, noise-free, static datasets.

SVR assumes a regular input grid, but our integrity dataset violates that assumption: each of the 25 late components has only 1-9 readings taken at irregular times. Resampling or imputation would be required which could be expensive or dilute information.

Even if the data was resampled, SVR only produces point forecasts. Estimating the probability of integrity falling below the 30% maintenance threshold would require additional calibration (e.g., Platt scaling), which is unreliable in extreme tails.

Additionally, SVR performance depends on tuning hyperparameters (C, ϵ and kernel width) requiring cross-validation which is impractical given our sparse data. Finally, whilst kernel methods interpolate well, the extrapolate poorly, making long-term forecasts from $t \leq 10$ to $t = 30$ unreliable.

By contrast, the hierarchical Bayesian model directly encodes the exponential-decay, pools strength across densely and sparsely sampled components, and outputs full predictive distributions - allowing direct estimation of failure probabilities.

6 Conclusions

This report has outlined and analysed a hierarchical Bayesian model in response to the clients three primary objectives.

The enhanced model accurately characterised the degradation rate, achieving an RMSE of 5.029 and 95% CI averaging $\pm 9.2\%$ from the function estimate. It outperformed both the simpler baseline Bayesian model (RMSE: 5.531) and black-box random forest (RMSE: 8.193). By pooling strength from well-populated data points, the hierarchical model was able to accurately model sparse data points and produce a probability that a component will need maintenance ($y_i < 30\%$). Finally the model identified X3 as the driving characteristic of decay, with X1 and X5 showing negligible impact.

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- Vehtari, A., Gelman, A., Simpson, D., Carpenter, B. and Bürkner, P.C., 2021. Rank-normalization, folding, and localization: An improved $\hat{r}$ for assessing convergence of mcmc (with discussion). *Bayesian analysis*, 16(2), pp.667–718.